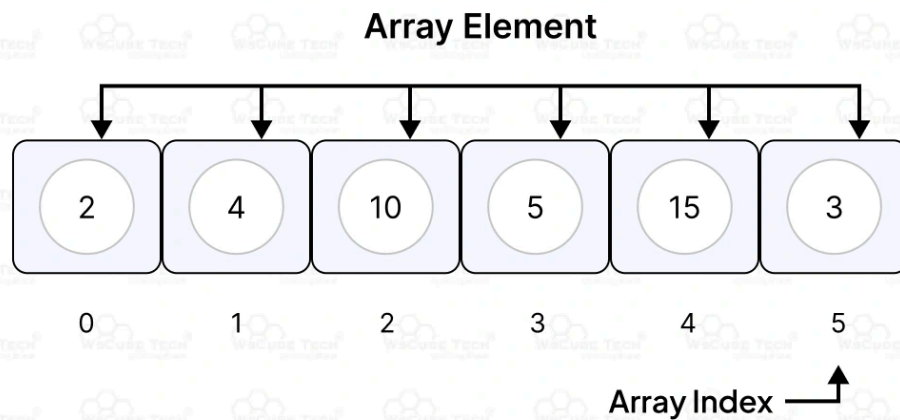


Array Data Structure



In JavaScript, an **array** is a global object used to store an ordered collection of multiple values under a single variable name. JavaScript arrays are dynamic (resizable) and can hold mixed data types, including numbers, strings, objects, or even other arrays. [\[1, 2, 3\]](#)

The **length property** returns or sets the total number of elements in a JavaScript array. It is a dynamic property that automatically updates when you add or remove elements

```
var colors = ["red", "green", "blue"];  
console.log(colors.length); // Output: 3
```

```
// Accessing the last element using length  
var lastColor = colors[colors.length - 1];  
console.log(lastColor); // Output: "blue"
```

Concatenation is the formal programming term for **joining things together end-to-end** to form a single item.

Think of it like linking train cars together to make a longer train. You are not changing the individual cars; you are just hooking them up in a specific sequence.

```
const text1 = "Coding";  
const text2 = "is";  
const text3 = "fun";  
  
// Joining them with spaces in between  
const sentence = text1 + " " + text2 + " " + text3 +  
"!";  
console.log(sentence);  
// Output: "Coding is fun!"
```